

Deep Hedging with Distributionally Robust, Curriculum, and Regime-Adaptive Neural Networks

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INTRODUCTION & MOTIVATION

Classical delta hedging assumes known dynamics and continuous rebalancing — both violated in practice. Deep hedging learns strategies directly from simulated paths, but three critical gaps remain:

- Distributional shift: trained on Heston → fails under regime change
- Training instability: CVaR → entropic jump destabilises optimisation
- Regime dependence: single network cannot adapt across market states

→ **W-DRO-T**: Wasserstein DRO with gradient-penalty dual
→ **3SCH**: Three-stage curriculum homotopy
→ **RSE**: Regime-switching ensemble with gating

Protocol: 80K Heston paths · N=30 steps · 10 seeds (42...942) · 10K bootstrap · Holm-Bonferroni ($\alpha=0.05$) · Cohen's d · SPY + NIFTY validation

Gap 1: Distributional Shift	→	W-DRO-T
Gap 2: Training Instability	→	3SCH
Gap 3: Regime Dependence	→	RSE

KEY RESULTS & STATISTICS

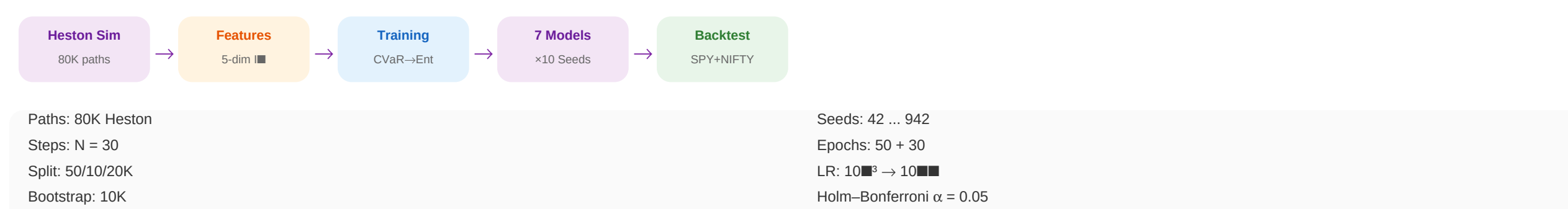
Simulated CVaR Comparison (10 Seeds)



Model	CVaR	95% CI	p vs LSTM	Cohen's d
RSE	3.109 ± 0.010	[3.104, 3.115]	< 0.0001	-7.41
LSTM	3.215 ± 0.015	[3.207, 3.223]	—	—
3SCH	3.219 ± 0.016	[3.210, 3.228]	0.58	+0.19
W-DRO-T	3.227 ± 0.011	[3.220, 3.234]	0.10	+0.60

RSE achieves lowest simulated CVaR (3.109 ± 0.010) with non-overlapping CI vs all baselines (p < 0.0001, |d| = 7.41), 10 seeds × 10K bootstrap.

EXPERIMENTAL PIPELINE



W-DRO-T Transformer	3SCH LSTM	RSE 3×LSTM
LSTM Baseline	Transformer Baseline	+2 More Ablations

MATHEMATICAL FORMULATION

Heston Stochastic-Volatility Model

$$dS_t = \mu S_t dt + \sqrt{V_t} S_t dW_t^S$$
$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_t^V$$
$$dW_t^S \cdot dW_t^V = \rho dt$$

Deep Hedging Objective

$$\mathbb{E}[\delta] = -Z + \mathbb{E}[\delta \Delta S] - c_{lc} \mathbb{E}[\delta] - \mathbb{E}[\delta \Delta S]$$
$$\theta^* = \arg\min_{\theta} p(\mathbb{E}[\delta])$$

Three Novel Risk Objectives

W-DRO-T — Wasserstein DRO dual:

$$\mathcal{L}_{DRO} = \mathbb{E}_P[\phi(X)] + \epsilon \cdot \mathbb{E}_P[\nabla_X \phi(X)], \quad \epsilon: 0 \rightarrow \epsilon_{\max}$$

3SCH — Three-stage curriculum homotopy:

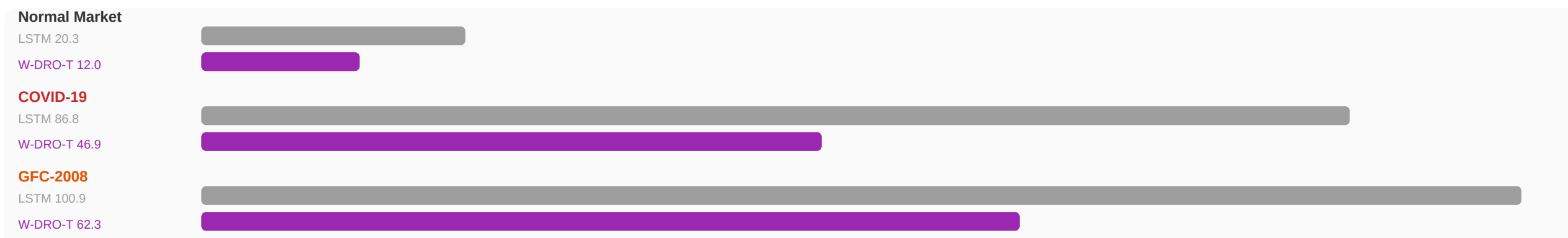
$$\mathcal{L}_{mix}(\alpha) = \alpha \cdot \text{CVaR} + (1-\alpha) \cdot p_{\lambda}, \quad \alpha: 0.8 \rightarrow 0.2$$

RSE — Regime-switching ensemble:

$$\hat{\delta} = \sum_{i=1}^K \nu_i \delta_i, \quad \nu_i = \frac{\exp(\text{softmax}(A_i))}{\sum_{j=1}^K \exp(\text{softmax}(A_j))}$$

REAL MARKET VALIDATION

SPY Crisis Stress Testing — CVaR



SPY (US, 5 bps cost)

- W-DRO-T dominates all scenarios
- Normal: -43% vs LSTM
- COVID-19: -46% vs LSTM
- GFC-2008: -38% vs LSTM
- Std P&L: 1.71 (vs 5.05, -66%)

NIFTY 50 (India, 18 bps cost)

- 3SCH wins normal: CVaR = 622
- Transformer wins COVID: 1644
- W-DRO-T wins GFC: 2350
- Higher vol (0.90) amplifies cost impact

KEY CONTRIBUTIONS

1. W-DRO-T: Wasserstein gradient-norm penalty with 2nd-order SDPA gradients
2. 3SCH: Mixed-loss homotopy via Berge's maximum theorem
3. RSE: Regime-gated ensemble with interpretable affinity $A \in \mathbb{R}^{K \times K}$
4. Statistical rigour: 10 seeds, 10K bootstrap, Holm-Bonferroni corrections
5. Market-specific deployment: US vs India evidence with crisis testing

METHOD OVERVIEW & NOVELTY

W-DRO-T Novelty

- Gradient-penalty dual avoids inner sup over Q
- ϵ -annealing stabilises early training
- Transformer with causal mask + 2nd-order SDPA

3SCH Novelty

- Mixed-loss convexity via Berge's maximum theorem
- Three-stage ϵ -annealing: warm-up, transition, fine-tune
- Zero-cost rollback to LSTM baseline

RSE Novelty

- Soft regime gating via learned affinity $A \in \mathbb{R}^{K \times K}$
- Regime features: realised vol, BIPV, returns, spread
- Variance reduction: $\text{Var}[\hat{\delta}] < \min_m \text{Var}[\delta_m]$

Statistical Rigour

- 10,000 bootstrap resamples for 95% CIs
- Paired t-tests with Holm-Bonferroni ($\alpha = 0.05$)
- Cohen's d effect size with magnitude classification

DEPLOYMENT & REGULATORY

Basel III/IV Capital Savings



Deployment Guidance

- US capital efficiency: W-DRO-T (-41% capital)
- India normal market: 3SCH (best NIFTY CVaR)
- India crisis: Transformer (best COVID)
- Regime uncertainty: RSE (CV = 0.30%)
- \$10B desk: \$32M/yr freed capital

Regulatory & Risk Compliance

- ✓ IFRS 9 / Ind-AS 109: 80–125% offset & $R^2 > 0.80$
- ✓ FRTB IMA: W-DRO-T lower P&L volatility
- ✓ SR 11-7: RSE affinity matrix interpretable
- ✓ DRO degrades slower (3.92× vs 4.28×)

FUTURE WORK & REFERENCES

Research Roadmap

- Online adaptation for non-stationary markets
- Multi-asset portfolio extension
- Market impact models integration
- Cross-market transfer learning
- Hybrid 3SCH-DRO architecture

Key References

- [1] Buehler et al. (2019) Deep Hedging, QF
- [2] Sinha et al. (2018) Certifying DRO, ICJR
- [3] Bengio et al. (2009) Curriculum Learning, ICML
- [4] Lakshminarayanan et al. (2017) Deep Ensembles
- [5] Rahmian & Mehrotra (2019) DRO Review

■ github.com/pratham-kailasiya/deep-hedging